What Should Reference Labels Buy? Estimation, Certification, and Repair in Conjunctive Evaluations

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Abstract

When a rubric task passes only if every criterion passes, checking an LLM judge against reference labels serves three goals: estimating accuracy, certifying outcomes, and repairing verdicts. We find that these gains can diverge sharply. In JudgmentBench, checking all 4,316 criteria on which two judges disagree corrects 1,217 criterion errors, yet immediate updating increases task disagreements from 244 to 245. To understand this outcome, we hold queried labels fixed and derive an exact accounting of new false passes prevented by certificate-gated updating and repairs delayed until certification. Under disagreement-first continuation, delayed repairs dominate the intermediate- and high-budget JudgmentBench comparison; at an 84% budget, the penalty is 6.0 or 3.9 disagreements per 100 matched outputs under two reference panels. Allocation has a different tradeoff: at a 20% budget, mixing certification with random sampling increases certification in all nine analysis cells, but repair gains are much larger in JudgmentBench than in RuVerBench. We trace this difference to which verdicts receive certificates and which repairs occur before certification. Finally, configurations with identical criterion confusion counts favor opposite update rules, showing why task associations are needed to plan verification. We use design-based estimators and finite-population identification bounds to distinguish statistical estimates from conclusions logically determined by queried labels and trusted summaries.

1 Introduction

Rubric-based evaluation breaks a complex output into several judgments. A model may satisfy a requested format, include the necessary evidence, and avoid a prohibited error. An evaluator then combines these judgments into a score or verdict. When success requires every retained condition to hold, one failed condition determines the outcome. APEX-Agents provides a practical example: its task-level Pass@1 requires all criteria to be satisfied (Vidgen et al., 2026). We observe a striking consequence in JudgmentBench: checking all 4,316 criteria on which GPT-5.4 and GPT-5.4-mini disagree corrects 1,217 criterion errors, yet task disagreements rise from 244 to 245. Four verdicts are repaired, but five correct rejections become false passes. The checks remove the judge's reasons for rejecting those outputs without reaching the criteria that actually fail. This example motivates a task-level analysis of what reference checking accomplishes.

A limited checking budget introduces a second problem. The same reference label can contribute to estimating the judge's overall accuracy, certify a particular task, or change an existing verdict. A checked failure may certify a task that was already correctly rejected. Correcting a mistaken rejection may repair another task before its remaining conditions are checked. These are distinct uses of evidence. An allocation that produces more certificates can therefore leave fewer, the same number, or more incorrect verdicts than another allocation at the same cost.

Research on multi-predicate screening, Boolean-function evaluation, active testing, and prediction-assisted inference supplies tools for choosing and interpreting checks (Krivosheev et al., 2018a; Gkenosis et al., 2022; Nguyen et al., 2018; Angelopoulos et al., 2023). We study how their different objectives interact in rubric evaluation. Our design separates *acquisition*, which chooses the labels to query, from *updating*, which decides

when to change a task verdict. Immediate updating substitutes every checked reference label and recomputes the conjunction. Certificate-gated updating changes the verdict only when the observed labels determine the whole task's outcome. Evaluating both rules on exactly the same acquired labels isolates the effect of the update rule.

We make two empirical contributions supported by exact accounting formulas. First, we connect task-level error configurations and query priorities to finite-budget update effects. We show how prevented false passes compete with delayed repairs, and find that the substantial high-budget delay in JudgmentBench survives the separately examined changes in reference and task composition; the small low-budget advantage is sensitive to those changes. Second, we trace allocation gains to certificates on initially correct and initially wrong verdicts, repairs made before certification, and newly introduced errors. This reveals why a fixed certification-plus-sampling allocation yields substantial repair gains in JudgmentBench but only small changes in RuVerBench. Our third research question tests the information needed for such comparisons: criterion summaries leave task associations unresolved, and restoring those associations narrows the conclusions compatible with the evidence.

The paper addresses three questions:

RQ1. For the same queried reference labels, how do immediate and certificate-gated updates change task verdicts, and which error structures and query orders determine their budget-dependent effects?

RQ2. How do sampling units and query allocations trade estimation precision against certification coverage and verdict repair, and when do certification gains translate into fewer residual disagreements?

RQ3. Which task-level conclusions are determined by criterion summaries and queried reference labels, what additional task associations are needed, and how does identification differ from statistical estimation?

Our evidence is a retrospective replay of released labels and saved judge predictions. We use those labels as specified references, and separately examine how changing the human reference changes the findings. This permits controlled comparisons without treating every recorded disagreement as an adjudicated judge error.

2 Setting and study design

2.1 Four evaluation targets

An evaluation frame contains N units. Unit i has k_i retained criteria, with binary reference labels Y_{ij} and judge predictions P_{ij}; one denotes PASS. Write $M = \sum_i k_i$, $Z_i = \prod_j Y_{ij}$, and $\widehat{Z}_i = \prod_j P_{ij}$. A unit is a task in RuVerBench and an output-by-rater annotation in the original JudgmentBench frame.

We distinguish criterion microaccuracy, $a = M^{-1} \sum_{ij} \mathbf{1}(P_{ij} = Y_{ij})$; the aggregate task pass rate, $\theta = N^{-1} \sum_i Z_i$; individual certification, which determines a particular Z_i; and verdict repair, which reduces disagreement between reported verdicts and Z_i. The first two are population quantities, while the last two concern individual outcomes. Counting erroneous criteria found by a policy is a further diagnostic, not a substitute for any of these targets.

A query reveals one reference bit. The query controller sees predictions, metadata, and previously queried labels; it cannot inspect future references. Unless additional information is explicitly supplied, unqueried bits can take either value. In this query-only model, one observed FAIL certifies failure and observing every bit as PASS certifies success. Classical Boolean evaluation provides the foundations for these certificates (Gkenosis et al., 2022; Ünlüyurt, 2004).

For a queried set Q, *eager* updating replaces P_{ij} by Y_{ij} on Q and recomputes the conjunction. *Certificate-gated* updating retains $\widehat{Z}_i$ until Q certifies an outcome, then reports that outcome. The gate therefore governs revisions; unresolved initial predictions remain visible. We count false passes (reported PASS, reference FAIL), false fails (reported FAIL, reference PASS), and their sum, denoted E.

2.2 Data and constructed scoring targets

RuVerBench supplies reference judgments on agentic coding (AC) and deep research (DR) criteria (Peng et al., 2026). We use saved predictions from four initial judge–domain runs and one additional GPT-5.4 low-reasoning

Table 1: Finite analysis frames. N counts tasks in RuVerBench and output-by-rater annotations in JudgmentBench; M counts criterion records. FP and FF compare the initial judge conjunction with the selected reference. The binary-only rows reuse the JudgmentBench annotations.

Frame / primary judge	N	M	Ref. PASS	FP	FF
DR Gemini 3.1 Pro	284	1,615	28	8	2
DR Qwen-plus	284	1,615	28	28	3
DR GPT-5.4 low	284	1,615	28	8	2
AC Qwen-plus	210	843	117	31	10
AC DeepSeek v4-pro	210	843	117	14	16
JB GPT-5.4, full	1,539	23,487	227	24	220
JB GPT-5.4-mini, full	1,539	23,487	227	36	223
JB GPT-5.4, binary-only	1,539	20,229	266	181	67
JB GPT-5.4-mini, binary-only	1,539	20,229	266	457	36

DR run. We retain the predictions saved for these five study runs. The conjunction covers the criteria retained in the released frame after upstream filtering.

JudgmentBench provides rubric annotations for legal outputs and saved GPT-5.4 and GPT-5.4-mini scores (Yang et al., 2026). Its native study concerns quality assessment by rubrics and preferences. We construct a strict all-pass target: every positive binary item must receive full credit, and every penalty occurrence count must be zero. The binary-only sensitivity removes penalty items. These choices define different success events. The original frame contains 1,539 annotations on 1,314 outputs from 30 base tasks, with 49 human raters. Table 1 describes all nine analysis cells; cells sharing outputs are not independent replications.

The full JudgmentBench target exposes a pronounced error imbalance: GPT-5.4 predicts only 31 PASS verdicts, of which seven agree with a reference PASS. Removing penalties raises predicted passes to 380 and reference passes to 266. Of the 220 initial full-target false fails, 156 reject only penalty conditions and 64 reject both item types. This is a useful stress test for update delay. It also means that its effect sizes should be interpreted alongside the changed target and the other frames.

2.3 Queries, budgets, and sensitivity designs

The policy study compares seven acquisition rules: uniform criterion sampling; short circuit with natural, randomized, or judge-failure-first within-task order; disagreement-only checking; disagreement checking followed by random continuation; and disagreement-prioritized short circuit. A disagreement flag records any available auxiliary prediction that differs from the primary. Short-circuit policies stop a task after a queried FAIL or a completed PASS certificate. The disagreement-first continuation rule visits flagged criteria task by task, then samples globally from the remaining criteria. Within that first phase, task order determines which disagreements are reached first.

We evaluate 101 budget caps from zero to the entire criterion frame, three task orders, and 31 randomized priority realizations, with a separate canonical seed-0 realization. Task orders are release order, ascending predicted-failure count, and random. A stopped policy can spend less than its cap, so comparisons report actual queries. The focal 20% allocation experiment compares criterion simple random sampling (SRS) with a fixed mix: up to half the budget goes to judge-first short circuit, followed by independent SRS from the unqueried remainder. The split is held fixed throughout the comparison.

Two additional designs examine stability. First, 206 repeatedly rated JudgmentBench outputs receive symmetric reference panels A and B, selected by an ID-hash rule. Both judge prediction vectors are held fixed. Crossed acquisition/evaluation references separate changing which labels are queried from re-reading the same query positions under another reference. Second, 399 whole-base-task resamples retain complete output, rater, and criterion blocks. Each replicate recalculates frame size and budget and reruns acquisition. Query-order variation, reference variation, and task-composition variation are reported separately. Appendices B–F provide the formulas, sampling procedures, and reconstruction constraints.

Table 2: Matched-query updating under disagreement-first continuation, release task order. Positive net differences mean more residual disagreements under gating. Expectations use the complete fixed reference; the last column averages 31 saved query orders. Counts refer to annotations, not independent base tasks.

JB GPT-5.4 target	Cap	Queries	$\mathbb{E}[A_b]$	$\mathbb{E}[D_b]$	$\mathbb{E}[D_b - A_b]$	Order mean
Full	20%	4,697	7.1	5.2	−2.0	−2.1
Full	84%	19,729	11.9	108.2	+96.3	+97.6
Binary-only	20%	4,046	28.2	36.4	+8.2	+8.3

3 RQ1: How updating changes task verdicts

3.1 A repair can expose another error

Consider a task with reference bits $(1, 0)$ and judge bits $(0, 1)$. Both conjunctions initially say FAIL. Checking the first criterion changes the eager bits to $(1, 1)$, producing a false PASS. Gating retains FAIL because the unchecked second criterion could still fail. The partial correction removed the reason for the original rejection without discovering the actual failure.

Now consider reference bits $(1, 1)$ with judge bits $(0, 1)$. Checking the first criterion repairs the eager verdict, but gating waits for the second check before reporting PASS. The same query structure that prevents an exposed false pass can postpone a valid repair. The empirical question is which of these events occurs more often at a given budget.

Let A_b count newly introduced false passes under eager updating and D_b count initial false fails that eager has repaired while gating still lacks a certificate.

Proposition 1 (Update accounting)**.** *For a fixed reference and the same queried set under both update rules,*

$$E_b^{\text{gate}} - E_b^{\text{eager}} = D_b - A_b. \tag{1}$$

Partitioning tasks by their initial and reference verdicts proves the identity (Appendix A.3). The net effect depends on when erroneous rejections are corrected and actual failures are discovered. Their positions within each task and the query order determine the length of that interval.

For uniform sampling of m criteria, an event requiring r particular bits to be observed and f other bits to remain unobserved has probability

$$h(M, m; r, f) = \frac{\binom{M-r-f}{m-r}}{\binom{M}{m}}, \tag{2}$$

with infeasible combinations assigned zero. For a susceptible initially correct FAIL, all predicted failures must be queried while every reference failure remains unqueried. For an initial false FAIL, delayed repair requires querying all predicted failures without completing the rubric. Summing these task events gives $\mathbb{E}[A_b]$ and $\mathbb{E}[D_b]$. Applying the same calculation within the active priority group gives exact expectations for disagreement-first continuation. The calculation is exact for the fixed reference and integrates over query randomization while retaining the observed dependence among criterion errors.

3.2 Delayed repairs dominate the high-budget JudgmentBench comparison

Under disagreement-first continuation, we find that delayed repairs dominate the high-budget JudgmentBench comparison (Table 2). The table separates analytical expectations from means over 31 query orders. In full JudgmentBench at a 20% cap, the expected prevention of new false passes slightly exceeds delayed repair. At 84%, the expected delay reaches 108.2 annotations while prevention remains 11.9. The net penalty is 96.3 disagreements. Both update rules converge to the reference at complete observation, so the penalty eventually returns to zero.

The task structure explains why the effect depends on the target. Full JudgmentBench has many initial false fails whose erroneous rejections can be corrected well before all passing conditions are checked. Binary-only has fewer false fails and many more initial false passes. Changing the target also changes rubric lengths,

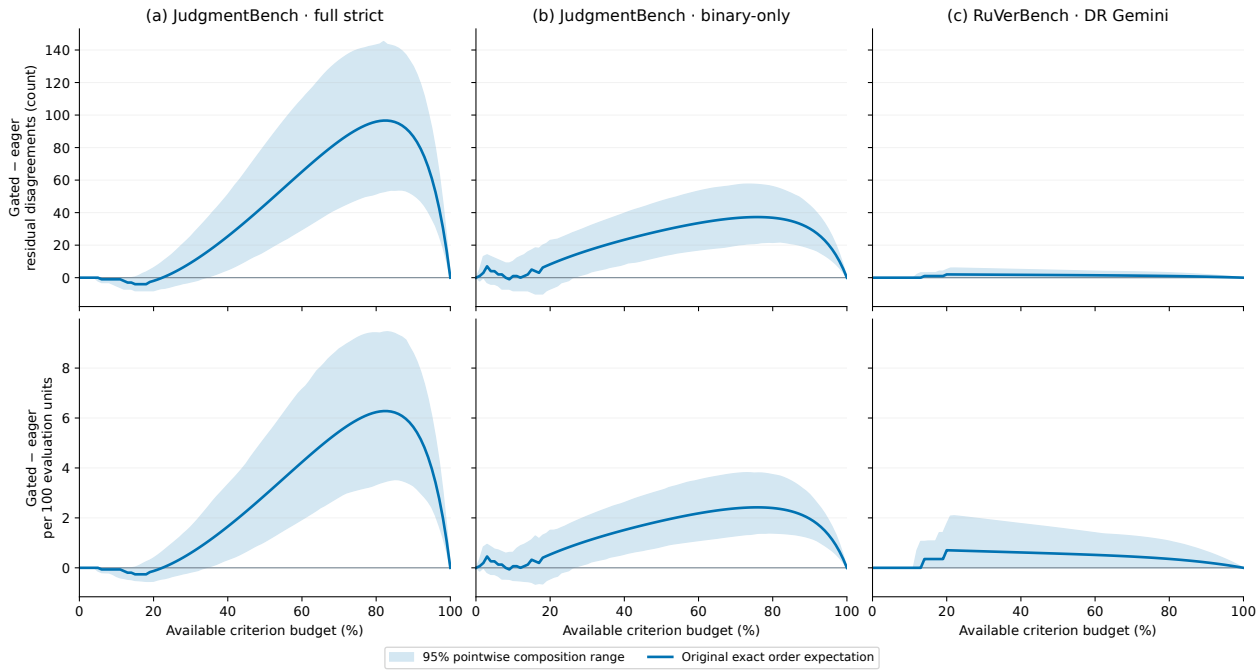


Figure 1: Complete-budget update comparisons in full JudgmentBench, binary-only JudgmentBench, and DR Gemini. The plotted disagreement-first continuation policy uses release task order. Original effects are analytical expectations; the shaded pointwise ranges summarize 399 whole-base-task resamples. Each replicate reruns the policy at its recalculated budget. These ranges describe task-composition sensitivity, separately from query-order or reference-panel variation. Both endpoint effects are exactly zero.

priorities, and absolute budgets, so this comparison describes a joint change in evaluation design. In DR Gemini there are only two initial false fails, sharply limiting the possible repair backlog. Moreover, a task whose every criterion is predicted FAIL has no such window: correcting all its rejected criteria completes its certificate at the same time.

Acquisition order constrains the size of the update effect.

Proposition 2 (Unresolved-task bound)**.** *Let $U(Q)$ count partly queried, uncertified units. With the same fixed reference and query set, $|E^{\mathrm{gate}}(Q) - E^{\mathrm{eager}}(Q)| \le U(Q)$. A serial policy that certifies each task before starting the next has $U(Q) \le 1$ at every prefix.*

Both rules agree on untouched and certified units, which proves the bound. Thus a large update effect requires checks dispersed across unfinished tasks. Figure 1 shows the complete-budget effects of disagreement-first continuation across three frames.

3.3 Which parts survive reference and composition changes?

In full JudgmentBench, the expected 20% contrast is -0.13 disagreements per 100 annotations, with a composition range of $[-0.54, 0.39]$. At 84%, it is $+6.26$, with range $[3.51, 9.44]$. The small low-budget advantage is unstable; the high-budget repair delay persists across the examined compositions.

The separate matched-output analysis gives the same qualitative distinction. On 206 fixed outputs, the 84% penalty is 6.0 or 3.9 disagreements per 100 outputs, depending on the reference panel. At 20%, both panels instead give a small positive effect. Moving from the original annotation frame to this matched-output frame also changes weighting, sample composition, and the saved judge anchor; it is not a pure reference replacement. Within the matched frame, reference choice can reverse a comparison: binary-only uniform sampling at 20% produces a mean difference of $+1$ under A and -1 under B.

These checks matter because the references themselves can disagree. In the repeated-annotation diagnostics,

Table 3: Allocation at a 20% criterion cap, release task order. Entries after queries are marginal medians across 31 orders. Width refers to a 95% conditional hypergeometric interval for criterion microaccuracy. The mix spends up to half its cap on certification, then samples the remainder. Pure certification uses judge-first short circuit; the dash indicates that no probability-sampling accuracy interval is reported for it.

Frame / primary	Allocation	Queries	Certified	Errors	Width (pp)
DR Gemini	SRS	323	131	8	5.5
	Fixed mix	323	169	9	7.6
	Pure certification	323	275	1	—
JB GPT-5.4, full	SRS	4,697	798	223	2.2
	Fixed mix	4,697	1,035	133	3.0
	Pure certification	4,697	1,512	15	—

32 of 54 initial false-fail annotations have another rater who also rejects the output, agreeing with the judge. Only 22 of 56 reference-PASS annotations are supported by every available peer rater. These are properties of the repeatedly rated subset, not estimates of which rater is correct. They motivate reporting the high-budget effect under both panels rather than treating the original 220 false fails as adjudicated model mistakes.

4 RQ2: What additional certificates buy

4.1 A common-budget comparison

We compare three uses of the same budget: random criterion sampling, a fixed certification-plus-sampling mix, and pure judge-first certification (Table 3). The mix increases certification and widens accuracy intervals in both focal frames. Pure certification produces still more resolved verdicts and fewer residual disagreements, but its adaptive queries do not supply the probability-sampling estimator used by the other two designs.

Across all nine cells, the mix increases certification in every paired query order. The mean change in residual disagreements ranges from about -2.3 to $+0.8$ across the five RuVerBench cells, compared with -91.5 to -58.7 across the four JudgmentBench cells. The focal DR Gemini comparison is the only positive mean contrast. The RuVerBench changes are small in absolute task counts. Below, we test task-composition sensitivity for the focal DR Gemini comparison and both strong-judge JudgmentBench targets.

We retain other acquisition controls to show that error discovery, certification, and repair can rank policies differently. At the DR 20% cap, judge-first short circuit finds a median of 14 criterion errors and certifies 275 tasks; disagreement-first continuation finds 83 errors and certifies 160. Both leave a median of one task error. The opposing discovery and certification ranking holds in all 31 paired orders. It also occurs for the weaker JudgmentBench judge, but in none of the orders for the stronger judge. When a weaker judge uses a stronger judge to flag disagreements, evaluating that stronger judge as the primary provides a direct comparison with simply using its verdicts from the outset.

4.2 Where the certificate gain goes

Let E_0 be the initial verdict disagreement count, and let C_W and C_R count certified units that were initially wrong and right. As in Proposition 1, D counts repairs made before certification and A counts introduced false passes.

Proposition 3 (Certification and repair accounting)**.** *At every query endpoint,*

$$C = C_R + C_W, \qquad E^{\text{eager}} = E_0 - C_W - D + A. \tag{3}$$

Certified initially wrong units are repaired; among uncertified units, the only repairs and new errors are D and A. For two allocations on the same frame,

$$\Delta C = \Delta C_R + \Delta C_W, \qquad \Delta E^{\text{eager}} = -\Delta C_W - \Delta D + \Delta A. \tag{4}$$

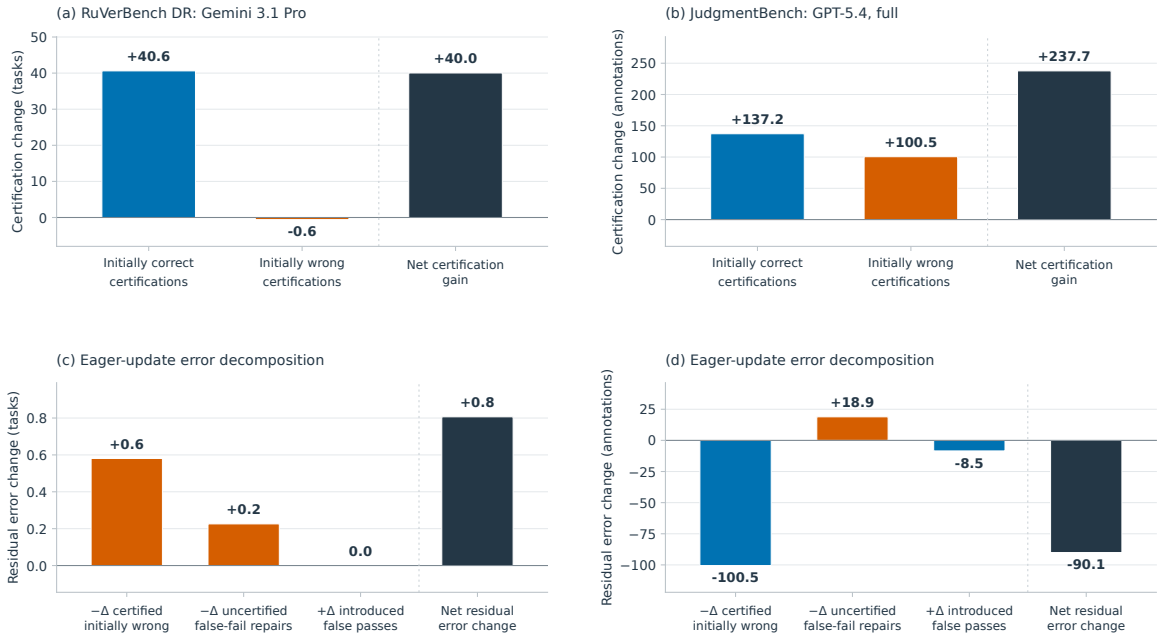


Figure 2: Allocation effects separated into where certificates are added and how residual disagreements change. Differences are fixed mix minus criterion SRS at a 20% budget, release task order, averaged across 31 paired orders at equal query counts. The columns show DR Gemini (284 tasks, 323 queries) and full JB GPT-5.4 (1,539 annotations, 4,697 queries). Components add before rounding. The DR error increase describes the original frame; its direction varies in the task-composition analysis.

A certificate on an already correct verdict adds evidence without repairing a mistake. Conversely, eager updating can repair a verdict before the task is certified. We use this distinction to locate the source of each allocation's gain.

In DR Gemini, the mix gains 40.0 certificates on average: 40.6 on initially correct tasks and −0.6 on initially wrong tasks. It also loses 0.2 uncertified repairs, leaving 0.8 more residual errors. The net certification gain therefore comes from verdicts that were already right. In JudgmentBench, certification increases by 237.7 overall and by 100.5 among initially wrong annotations. The mix loses 18.9 uncertified repairs but introduces 8.5 fewer new false passes, leaving 90.1 fewer disagreements. Figure 2 displays the additive paired components.

The pattern persists differently under sensitivity checks. In JudgmentBench, both full and binary-only allocation comparisons reduce error in every one of the 399 task resamples. On the matched 206-output frame, all four acquisition/evaluation-reference combinations improve mean certification and repair at 620 queried labels. The matched-reference gains are +27.6 certificates and −11.7 disagreements under A, and +32.1 and −8.7 under B. Re-reading A's query positions under B gives +14.9 and −4.5; re-reading B's positions under A gives +13.6 and −7.3. A few individual orders in the crossed comparisons increase disagreement, even though their paired means improve.

DR gains certificates in every resample, but its repair contrast has 196 reductions, 198 increases, and five ties. We find stable certification gains in this DR Gemini comparison, while its small repair contrast changes with task composition.

4.3 Preserving estimation while targeting certification

The fixed mix also provides a transparent sampling estimator. If its first phase observes set C, the remainder contains $R = M - |C|$ bits. Draw n of those bits by SRS and write S for the sample. With $a_u = \mathbf{1}(P_u = Y_u)$,

$$\widehat{a} = \frac{\sum_{u \in C} a_u + (R/n) \sum_{u \in S} a_u}{M}. \tag{5}$$

Conditional on the first phase, this estimator is unbiased because the second phase is an independent probability sample. We invert hypergeometric tails for the unknown remainder agreement count and add the known first-phase total. This yields a fixed-budget conditional interval. Its justification follows from the design, in the tradition of finite-population sampling (Horvitz and Thompson, 1952), rather than from the observed coverage in 31 repeats.

Alongside SRS, we compare proportional stratification and pilot-based Neyman allocation using primary predicted PASS/FAIL strata. Pilot labels count toward the budget. Known pilot totals are added to estimated remaining stratum totals; hypergeometric count bounds are combined with Bonferroni error allocation. These controls isolate a conventional design-based alternative alongside the prediction-powered methods developed in prior work (Angelopoulos et al., 2023; Fisch et al., 2024; Kim, 2026).

Conservative interval width and empirical precision tell different stories. At the full-JudgmentBench 20% cap, mean guaranteed widths are 2.21 pp for SRS, 3.47 for proportional stratification, and 3.74 for pilot Neyman. Yet proportional stratification reduces empirical RMSE in AC DeepSeek from 2.35 to 1.40 pp and in DR Gemini from 1.37 to 1.19 pp. Full JudgmentBench instead changes from 0.50 to 0.57 pp. Figure 3 reports both width and RMSE for all nine cells, retaining improvements and contrary outcomes.

Estimating θ calls for another sampling unit. Select a fixed SRS of tasks and certify their outcomes, either by reading every rubric item or by stopping at the first failure. Both procedures observe the same sampled Z_i and give the same pass-rate estimator. For 57 sampled DR tasks, median query cost falls from 319 to 67; for 308 JudgmentBench annotations, it falls from 4,690 to 1,002. Their exact design RMSEs for task pass rate are 3.54 and 1.81 pp. This fixed-unit-count design has variable label cost; it answers a different resource question from Table 3.

5 RQ3: What criterion summaries leave unresolved

The update mechanism depends on which failures co-occur within a task and where they lie in the query order. Criterion confusion counts do not retain these associations. We test their sufficiency by shuffling references within task-length × primary-prediction × criterion-category strata, preserving the stratum confusion counts, task lengths, predictions, and observable priorities.

Across 64 such configurations, the full-JudgmentBench uniform-sampling effect at 20% changes from +12.25 disagreements in the original frame to values between −9.76 and −1.48. We observe the opposite preferred update rule under equal error costs in every configuration, despite identical criterion summaries. Across the nine cells, two policies, and five caps, 44 of 90 conditions admit an opposite-sign effect. These are counterexamples to information sufficiency, not estimates of a reversal frequency in new settings. With every stratum's confusion counts preserved, reference passes fall from 227 to between 108 and 132 across these 64 configurations. These changes jointly alter task success and error structure; our comparison tests the sufficiency of the summaries.

We next hold the reference frame fixed and vary what information is disclosed. In the query-only model, let n_P and n_F denote certified PASS and FAIL counts. Unrestricted completions give the sharp endpoint range

$$\theta \in \left[\frac{n_P}{N}, 1 - \frac{n_F}{N}\right]. \tag{6}$$

Each unresolved task can independently be completed to PASS or FAIL, so the endpoints are attainable. More certificates narrow this range. A probability sample can nevertheless give a useful statistical estimate of θ while many individual tasks remain unresolved: logical identification and sampling precision answer different questions (Rao and Callison-Burch, 2026; Chen et al., 2026).

Additional trusted summaries change the feasible set. Our disclosure replay begins with reference-containing confusion margins grouped by task length and prediction, refines them by criterion category, and then restores labels linked to whole tasks. Binary mixed-integer optimization computes attainable minimum and maximum pass totals. For DR Gemini, the initial margins leave the pass rate between 4.23% and 16.90%, a width of 12.68 pp. These margins comprise 38 disclosed count cells, not 38 new reference queries. Full link restoration recovers the observed pass rate.

Appendix F gives a complete two-task example of how a trusted total and linked observations jointly identify the pass rate. Aggregate identification can still leave individual verdicts unresolved.

The practical implication is to report the information that supports the intended conclusion. Linked criterion records support analysis of update and certification behavior; criterion-level confusion summaries alone generally do not. RuVerBench already releases linked records, which make this replay possible. Preserving these links in downstream validation reports lets evaluators examine consequences that aggregate agreement conceals.

6 Related work

Multi-predicate screening studies condition-level crowd queries, posterior stopping, and downstream classification loss (Krivosheev et al., 2018a,b, 2020). Sequential system testing and stochastic Boolean-function evaluation formalize how costly component checks determine an outcome (Ünlüyurt, 2004; Gkenosis et al., 2022; Ghuge et al., 2026). Structured agent evaluation also uses node-level checks and short circuit (Gou et al., 2025). Our logical certificates use these foundations. The empirical extension is to compare their role in repairing an already available judge verdict on matched queried sets.

Selective correction and local error propagation have close precedents. ActiveClean separates sample selection from model updating and shows why mixed clean and dirty data can distort learning (Krishnan et al., 2016). Query-Guided Verification traces AND/OR query provenance, including cases where lowering a component's error probability raises a worst-case output score (Schreiber and Amsterdamer, 2026). Its maximum-error score uses independent verification errors and differs from our realized FP/FF counts after exact bit replacement. Correction-and-corruption analysis also compares selective and unconditional application on fixed candidates and selects applications using estimated correction and corruption rates (Reitich, 2026). Our contribution rests on the event decomposition, quantified budget curves, and their observed stability in rubric frames, building on these correction principles.

Failure discovery and performance estimation have long motivated different sample choices. DeepEST combines failure-oriented selection with population estimation, while recent multi-objective test selection compares discovery, estimation, and retraining (Guerriero et al., 2021; Zhang et al., 2026). Active Testing and prediction-assisted inference use partial trusted labels to estimate aggregate performance (Nguyen et al., 2018; Chaganty et al., 2018; Angelopoulos et al., 2023). StratPPI studies stratified allocation, and Kim uses a two-phase doubly robust design and pilot information to plan human-review requirements (Fisch et al., 2024; Kim, 2026). We add the joint accounting of certification and verdict repair, while using conventional sampling controls for estimation.

Different guarantees govern different decisions. Trust or Escalate calibrates acceptance and escalation of pairwise judgments to control disagreement risk among accepted judgments under its calibration assumptions (Jung et al., 2025); audit-efficient best-agent identification targets system selection (Ao et al., 2026). A logical certificate instead resolves a particular conjunction from observed reference bits. Evaluation-protocol and partial-identification work further distinguishes the conclusions licensed by aggregated information (Rao and Callison-Burch, 2026; Chen et al., 2026).

Finally, human uncertainty affects automatic-evaluator assessment (Elangovan et al., 2025). Physician disagreement in HealthBench has been studied through variance decomposition and prediction (Borgohain and Mariathas, 2026). Our reference sensitivity carries this concern through the checking protocol: it fixes both judges, changes the reference panel, and separates re-reading a fixed query set from rerunning acquisition. The resulting differences concern protocol conclusions as well as initial agreement.

7 Discussion and limitations

Our comparisons suggest three practical choices. For criterion accuracy, use probability sampling, with stratification or a certification phase when those serve the evaluation's goals. For individual certificates and logical pass-rate bounds, concentrate checks within tasks; judge-failure-first short circuit resolves many verdicts in the examined frames. For repair, choose an error tradeoff explicitly and consider how many tasks the acquisition policy leaves unfinished. A serial task-completing policy limits the eager–gated difference to one verdict, whereas dispersed checks can make the update rule consequential across many tasks.

For false-pass and false-fail costs c_{FP} and c_{FF}, the expected gated-minus-eager loss is $c_{FF}\mathbb{E}[D] - c_{FP}\mathbb{E}[A]$. When $c_{FF} > 0$ and $\mathbb{E}[A] > 0$, gating has lower expected loss exactly when $c_{FP}/c_{FF} > \mathbb{E}[D]/\mathbb{E}[A]$. Using the unrounded expectations underlying Table 2, the break-even ratio is approximately 0.725 at a 20% budget and 9.1 at 84%. At the 84% budget in this comparison, waiting for certificates pays off only when a false pass costs more than roughly nine false fails. These thresholds describe the saved frame. Prospective selection would require affordable estimates of the relevant error structures, a charged pilot budget, and confirmation on unused data.

The study has four limits. First, all repairs and certificates refer to released labels. We do not adjudicate the raters, and a rubric certificate does not establish complete occupational competence or correctness under requirements outside the rubric. Second, our all-pass targets are constructed from two data families. The large full-JudgmentBench update effect occurs in a FAIL-dominated judge regime; the binary-only and DR comparisons show how its magnitude changes. Third, the experiments are retrospective. Repeated query orders do not create new tasks, and resampling 30 curated base tasks assesses composition sensitivity without making them representative of every future environment. Reference and task-composition sensitivities are separate analyses, not a joint uncertainty interval. Fourth, query counts measure revealed bits, not expert time or money. They omit the cost of obtaining auxiliary predictions and resolving reference disagreement.

The complete-reference calculations are explanations of the observed frames. A new environment may have different task lengths, pass rates, error overlap, or query priorities. A useful extension would test a prespecified prediction of protocol performance on unused tasks or on a native all-pass benchmark. Such a study would address transfer of the mechanism; a claim about expert labor would additionally need measured annotation costs. The present paper supplies the decomposition and reproducible comparisons with which to formulate those tests.

8 Conclusion

A reference check can improve an estimate, establish an outcome, or repair a verdict, and these gains have different information requirements. Matched-query comparisons show that the cost of waiting for a certificate depends on the interval between correcting erroneous rejections and completing the task. Allocation comparisons show that additional certificates improve repair only through the tasks and error events they resolve. Reference and task-composition checks distinguish persistent effects from small, unstable contrasts. Reference labels should buy the evidence the evaluation needs: probability samples for population estimates, completed checks for individual certificates, and timely correction for verdict repair. Choosing among these objectives also means choosing where to query and when to update.

Data, code, and AI assistance

Data, saved predictions, analysis code, figures, and manuscript build instructions are available at https://github.com/Itsxuchen/RefEval. Reproduction requires no model API calls. The package preserves upstream attribution and data terms and redistributes compact numerical records. JudgmentBench judge predictions are source-published; RuVerBench predictions come from our saved runs.

Codex and Claude assisted with code, analysis checks, literature searches, and drafting. We checked generated material against source papers, code, and saved data. The named human author is responsible for the manuscript.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023. doi: 10.1126/science.adi6000. URL https://arxiv.org/abs/2301.09633v4.

Ruicheng Ao, Hongyu Chen, Siyang Gao, Hanwei Li, and David Simchi-Levi. Best arm identification with LLM judges and limited human audits, 2026. URL https://arxiv.org/abs/2601.21471v1.

Satya Borgohain and Roy Mariathas. Decomposing physician disagreement in HealthBench, 2026. URL https://arxiv.org/abs/2602.22758v2.

Arun Tejasvi Chaganty, Stephen Mussmann, and Percy Liang. The price of debiasing automatic metrics in natural language evaluation. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 643–653, 2018. doi: 10.18653/v1/P18-1060. URL https://aclanthology.org/P18-1060/.

Xiaohong Chen, Ashesh Rambachan, and Elie Tamer. Partial identification from LLM prompts, 2026. URL https://arxiv.org/abs/2606.15031v2.

Aparna Elangovan, Lei Xu, Jongwoo Ko, Mahsa Elyasi, Ling Liu, Sravan Babu Bodapati, and Dan Roth. Beyond correlation: The impact of human uncertainty in measuring the effectiveness of automatic evaluation and LLM-as-a-judge. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/8798321486948322be2b4d658744ba72-Abstract-Conference.html.

Adam Fisch, Joshua Maynez, R. Alex Hofer, Bhuwan Dhingra, Amir Globerson, and William W. Cohen. Stratified prediction-powered inference for effective hybrid evaluation of language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/c9fcd02e6445c7dfbad6986abee53d0d-Abstract-Conference.html.

Rohan Ghuge, Jai Moondra, and Mohit Singh. Stochastic function certification with correlations, 2026. URL https://arxiv.org/abs/2604.02611v1.

Dimitrios Gkenosis, Nathaniel Grammel, Lisa Hellerstein, and Devorah Kletenik. The stochastic boolean function evaluation problem for symmetric boolean functions. *Discrete Applied Mathematics*, 309:269–277, 2022. doi: 10.1016/j.dam.2021.12.001. URL https://arxiv.org/abs/2111.08793v3.

Boyu Gou, Zanming Huang, Yuting Ning, Yu Gu, Michael Lin, Weijian Qi, Andrei Kopanev, Botao Yu, Bernal Jiménez Gutiérrez, Yiheng Shu, Chan Hee Song, Jiaman Wu, Shijie Chen, Hanane Nour Moussa, Tianshu Zhang, Jian Xie, Yifei Li, Tianci Xue, Zeyi Liao, Kai Zhang, Boyuan Zheng, Zhaowei Cai, Viktor Rozgic, Morteza Ziyadi, Huan Sun, and Yu Su. Mind2Web 2: Evaluating agentic search with agent-as-a-judge, 2025. URL https://arxiv.org/abs/2506.21506v2.

Antonio Guerriero, Roberto Pietrantuono, and Stefano Russo. Operation is the hardest teacher: Estimating DNN accuracy looking for mispredictions. In *2021 IEEE/ACM 43rd International Conference on Software Engineering (ICSE)*, pages 348–358, 2021. doi: 10.1109/ICSE43902.2021.00042. URL https://arxiv.org/abs/2102.04287v1.

D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952. doi: 10.1080/01621459.1952.10483446. URL https://doi.org/10.1080/01621459.1952.10483446.

Jaehun Jung, Faeze Brahman, and Yejin Choi. Trust or escalate: LLM judges with provable guarantees for human agreement. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/08dabd5345b37fffcbe335bd578b15a0-Abstract-Conference.html.

Jane Paik Kim. Augmenting human evaluation with LLM judges: How many human reviews do you need?, 2026. URL https://arxiv.org/abs/2605.16354v1.

Sanjay Krishnan, Jiannan Wang, Eugene Wu, Michael J. Franklin, and Ken Goldberg. ActiveClean: Interactive data cleaning for statistical modeling. *Proceedings of the VLDB Endowment*, 9(12):948–959, 2016. doi: 10.14778/2994509.2994514. URL https://activeclean.github.io/files/activeclean-vldb16.pdf.

Evgeny Krivosheev, Fabio Casati, Marcos Baez, and Boualem Benatallah. Combining crowd and machines for multi-predicate item screening. *Proceedings of the ACM on Human-Computer Interaction*, 2(CSCW):97, 2018a. doi: 10.1145/3274366. URL https://marcosbaez.com/assets/pdf/krivosheev2018combining.pdf.

Evgeny Krivosheev, Fabio Casati, and Boualem Benatallah. Crowd-based multi-predicate screening of papers in literature reviews. In *Proceedings of the 2018 World Wide Web Conference*, 2018b. doi: 10.1145/3178876.3186036. URL https://arxiv.org/abs/1803.09814v1.

Evgeny Krivosheev, Burcu Sayin, Alessandro Bozzon, and Zoltán Szlávik. Active learning from crowd in document screening. In *Proceedings of the Crowd Science Workshop: Remoteness, Fairness, and Mechanisms as Challenges of Data Supply by Humans for Automation*, volume 2736 of *CEUR Workshop Proceedings*, pages 19–25, 2020. URL https://ceur-ws.org/Vol-2736/paper4.pdf.

Phuc Nguyen, Deva Ramanan, and Charless Fowlkes. Active testing: An efficient and robust framework for estimating accuracy. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3759–3768, 2018. URL https://proceedings.mlr.press/v80/nguyen18d.html.

Yangda Peng, Yunjia Qi, Haotian Xia, Guanzhong He, Xintong Shi, Richeng Xuan, Songyuanyi Lu, Yixian Liu, Zhichao Hu, Yuhong Liu, and Hao Peng. Can LLM-as-a-judge reliably verify rubrics in agentic scenarios?, 2026. URL https://arxiv.org/abs/2606.29920v2.

Delip Rao and Chris Callison-Burch. Agreement metrics for LLM-as-judge evaluation: What to report and why, 2026. URL https://arxiv.org/abs/2606.00093v2.

Fernando Reitich. Correction and corruption: A two-rate view of error flow in LLM protocols, 2026. URL https://arxiv.org/abs/2604.18245v3.

Ran Schreiber and Yael Amsterdamer. Query-guided analysis and mitigation of data verification errors. In *2026 IEEE 42nd International Conference on Data Engineering (ICDE)*, pages 1994–2007, 2026. doi: 10.1109/ICDE65706.2026.00151. URL https://arxiv.org/abs/2603.08612v2.

Tonguç Ünlüyurt. Sequential testing of complex systems: a review. *Discrete Applied Mathematics*, 142(1–3):189–205, 2004. doi: 10.1016/j.dam.2002.08.001. URL https://www.cs.umd.edu/projects/reucaar/surveyeval.pdf.

Bertie Vidgen, Austin Mann, Abby Fennelly, John Wright Stanly, Lucas Rothman, Marco Burstein, Julien Benchek, David Ostrofsky, Anirudh Ravichandran, Debnil Sur, Neel Venugopal, Alannah Hsia, Isaac Robinson, Calix Huang, Olivia Varones, Daniyal Khan, Michael Haines, Austin Bridges, Jesse Boyle, Koby Twist, Zach Richards, Chirag Mahapatra, Brendan Foody, and Osvald Nitski. APEX–agents, 2026. URL https://arxiv.org/abs/2601.14242v3.

Russell Yang, Ruishi Chen, Pierce Kelaita, Riya Ranjan, Sibo Ma, Charles Dickens, Matthew Guillod, Megan Ma, and Julian Nyarko. JudgmentBench: Comparing rubric and preference evaluation for quality assessment, 2026. URL https://arxiv.org/abs/2605.25240v3.

Jingyu Zhang, Fan Wang, Jacky Keung, Yihan Liao, Yan Xiao, and Lei Ma. Empirical insights of test selection metrics under multiple testing objectives and distribution shifts. *Proceedings of the ACM on Software Engineering*, 3(FSE):FSE058, 2026. doi: 10.1145/3797086. URL https://arxiv.org/abs/2604.23342v1.

A Finite reference frames and verdict accounting

Let $\mathcal{J}_i$ be the nonempty set of criterion slots for evaluation unit i, with $k_i = |\mathcal{J}_i|$, N units, and $M = \sum_i k_i$ slots. Slots belong to exactly one unit. The released reference and primary prediction for slot u are $Y_u, P_u \in \{0, 1\}$, where one denotes PASS. Throughout this appendix they are fixed finite-population values. Define

$$Z_i = \prod_{u \in \mathcal{J}_i} Y_u, \qquad \widehat{Z}_i^0 = \prod_{u \in \mathcal{J}_i} P_u, \qquad \theta = \frac{1}{N} \sum_i Z_i. \tag{7}$$

Criterion microagreement is the distinct target $a = M^{-1} \sum_u \mathbf{1}\{Y_u = P_u\}$. An error in the following equations means disagreement with this particular reference. No equation identifies which of two disagreeing human raters is substantively correct.

For JudgmentBench full strict, a positive binary item passes when its original value is one, and a negative-weight occurrence-count item passes when its original count is zero. The loader checks that their conjunction equals attaining the rubric's maximum score. Binary-only retains positive binary items and removes occurrence-count items. This changes both the success event and the criterion population. The original unit is an output-by-rater annotation; the alternative-reference analysis instead weights distinct outputs equally. RuVerBench units use the criteria retained in its release.

A.1 Certificates and update rules

Let $Q \subseteq \bigcup_i \mathcal{J}_i$ be the queried slots and $Q_i = Q \cap \mathcal{J}_i$. In the query-only information model, a FAIL certificate is an observed reference failure, and a PASS certificate requires observing every slot as PASS. Write

$$C_i^- = \mathbf{1}\{\exists u \in Q_i : Y_u = 0\}, \tag{8}$$

$$C_i^+ = \mathbf{1}\{Q_i = \mathcal{J}_i,\ Z_i = 1\}, \qquad C_i = C_i^- + C_i^+. \tag{9}$$

Eager updating substitutes every queried reference value immediately:

$$\widehat{Z}_i^E(Q) = \prod_{u \in \mathcal{J}_i} \big(\mathbf{1}\{u \in Q\} Y_u + \mathbf{1}\{u \notin Q\} P_u\big). \tag{10}$$

Certificate-gated updating uses Z_i once certified and otherwise retains $\widehat{Z}_i^0$:

$$\widehat{Z}_i^G(Q) = C_i Z_i + (1 - C_i)\widehat{Z}_i^0. \tag{11}$$

Writing Z_i here is shorthand for the value logically determined by a certificate, not access to an unqueried reference. The implementation obtains that value from the queried bits. Gating can retain an uncertified initial PASS; it is not a protocol that publishes only certified outcomes.

A.2 Acquisition implementations

The seven acquisition rules use a common slot map. Uniform criterion sampling follows a global random priority. Three serial short-circuit rules visit tasks in the stated order and visit their criteria in natural slot order, random order, or predicted-FAIL-first order, respectively. They stop the current task at its first observed reference failure or after completing its rubric. Disagreement-only visits flagged criteria within each task and stops after the flagged pool is exhausted; it can leave budget unused. Disagreement-then-random uses that same flagged prefix, followed by a global random ordering of all unflagged criteria. Disagreement-prioritized short circuit visits, within a task, flagged slots first, then unflagged predicted FAIL slots, then the remaining predicted PASS slots; its stopping rule is the same serial certification rule.

For each DR primary, the auxiliaries are the other two saved DR judges among Gemini 3.1 Pro, Qwen-plus and GPT-5.4 low. Each AC primary has the other saved AC judge as its auxiliary: Qwen-plus or DeepSeek v4-pro. JudgmentBench pairs GPT-5.4 with GPT-5.4-mini within the same scoring target. The flag is the union of disagreements with available auxiliaries. Their predictions are treated as available signals; their acquisition cost is not included in the reference-bit budget.

The three task orders are release-relative order, ascending primary predicted-failure count, and a random task order. Seeds 1–31 randomize ties within the declared priorities. A separate canonical seed-zero realization uses natural slot order for priority ties while retaining random priorities for explicitly random policies and the random task order. Purpose-specific streams derive from root seed 20260913, the replicate seed and a SHA256 hash of the stream's name. Release-relative order refers to the loader's retained unit ordering, not a randomized sample of deployment tasks. The analytical mechanism calculations below hold this order fixed.

A.3 Exact event decomposition

Define predicted-failure slots $F_i = \{u \in \mathcal{J}_i : P_u = 0\}$ and reference-failure slots $T_i = \{u \in \mathcal{J}_i : Y_u = 0\}$. An initially correct FAIL is susceptible to a new false PASS only if F_i and T_i are nonempty and disjoint. Its new-error event is

$$A_i(Q) = \mathbf{1}\{F_i \neq \varnothing,\ T_i \neq \varnothing, F_i \cap T_i = \varnothing,\ F_i \subseteq Q, T_i \cap Q = \varnothing\}. \tag{12}$$

An initially false FAIL has $T_i = \varnothing$ and $F_i \neq \varnothing$. Eager updating repairs it once all its predicted failures have been checked. Gating delays that repair until every criterion has been checked. The delayed-repair event is therefore

$$D_i(Q) = \mathbf{1}\{T_i = \varnothing,\ F_i \neq \varnothing, F_i \subseteq Q,\ \mathcal{J}_i \not\subseteq Q\}. \tag{13}$$

Let $A(Q) = \sum_i A_i(Q)$, $D(Q) = \sum_i D_i(Q)$, and $E_r(Q) = \sum_i \mathbf{1}\{\widehat{Z}_i^r(Q) \neq Z_i\}$ for $r \in \{E, G\}$. Then

$$E_G(Q) - E_E(Q) = D(Q) - A(Q). \tag{14}$$

Derivation. Initially correct PASS units have every prediction and reference equal to one, so either update remains correct. An initially false PASS is repaired exactly when a reference failure is queried, which also certifies FAIL; both updates therefore agree. For an initially correct FAIL, an unqueried predicted failure or a queried reference failure prevents eager PASS. Eager can introduce a false PASS precisely in (12), while gating retains the correct FAIL. For an initially false FAIL, the two rules disagree precisely in (13). Summing these four exhaustive initial classes proves (14) and Proposition 1. In particular, correct reference substitutions cannot introduce a false FAIL under conjunction.

If false PASS and false FAIL have fixed nonnegative costs c_{FP} and c_{FF}, their endpoint loss difference is $c_{\mathrm{FF}}D - c_{\mathrm{FP}}A$. Thus equal-weight error comparisons are one particular loss choice. Along a growing query transcript, gated error cannot increase: every changed verdict is certified relative to the fixed reference. Its difference from eager error can increase because eager may repair false FAILs earlier. At zero queries and at a census the difference is zero.

Both rules agree on untouched and certified units. If $U(Q)$ units have been partially queried but remain uncertified, then $|E_G(Q) - E_E(Q)| \leq U(Q)$. A serial short-circuit policy that completes or certifies a task before starting the next has $U(Q) \leq 1$ at every prefix. This explains why changing the update rule has little room to change the endpoint under that acquisition structure, even when dispersed querying has a large effect.

B Finite-budget expectations over query order

For uniform sampling of h slots without replacement from a group of size G, the probability of including r specified slots and excluding f other specified slots is

$$H(G, h; r, f) = \frac{\binom{G-r-f}{h-r}}{\binom{G}{h}}. \tag{15}$$

The specified inclusion and exclusion sets are disjoint. We set the probability to zero when $r + f > G$, $r > h$, or $f > G - h$; in the empty case $H(0, 0; 0, 0) = 1$. Equation (15) follows by choosing the other $h - r$ observed slots from the $G - r - f$ unrestricted slots. It is a sampling calculation and assumes no independence of reference errors.

For a uniform m-of-M criterion query set, the two expectations are

$$\mathbb{E}[A(Q)] = \sum_{i:\, F_i, T_i \neq \varnothing,\, F_i \cap T_i = \varnothing} H(M, m; |F_i|, |T_i|), \tag{16}$$

$$\mathbb{E}[D(Q)] = \sum_{i:\, T_i = \varnothing,\, F_i \neq \varnothing} \{H(M, m; |F_i|, 0) - H(M, m; k_i, 0)\}. \tag{17}$$

The complete-rubric event is contained in the event of observing every predicted failure, which justifies the subtraction in the second line. Linearity of expectation gives the expected update difference even though events for different tasks are dependent under a fixed global budget.

B.1 Prioritized groups

The same calculation applies to a predetermined sequence of disjoint query groups $\mathcal{G}_1, \ldots, \mathcal{G}_L$. Every slot in an earlier group is queried before any slot in a later group, and order within each group is uniformly random. At an interior cap m, let ℓ be the group satisfying

$$\sum_{g<\ell} |\mathcal{G}_g| \leq m < \sum_{g\leq\ell} |\mathcal{G}_g|, \qquad h = m - \sum_{g<\ell} |\mathcal{G}_g|.$$

For a required set R and forbidden set S, the event $R \subseteq Q$, $S \cap Q = \varnothing$ has probability zero if a required slot lies after ℓ, a forbidden slot lies before ℓ, or the two sets overlap. Otherwise its probability is

$$H\big(|\mathcal{G}_\ell|, h; |R \cap \mathcal{G}_\ell|, |S \cap \mathcal{G}_\ell|\big). \tag{18}$$

At a complete group boundary, the next group has $h = 0$. At a census the event is determined directly by the complete query set. Substituting this event probability for each term in (17) gives the grouped-order expectations.

For disagreement-then-random, a criterion is flagged when any available auxiliary judge disagrees with the primary prediction. Each task's flagged criteria form one group, tasks follow release order, and the final group contains all unflagged criteria. The flagged tier is therefore not a single global random sample. These groups depend on saved predictions and task order, not hidden references. The analytical grid covers uniform criterion and disagreement-then-random querying at 101 shares from zero to one. It does not apply the predetermined-group formula to adaptive short-circuit paths. Budget caps use the implementation's nearest-integer rule $m = \text{round}(bM)$, with ties to an even integer.

All expectations condition on the full reference, predictions and priority groups. They explain the observed frames rather than provide a free prospective policy selector. The 31 random-order means provide a Monte Carlo check against these expectations; their standard error is the sample standard deviation divided by $\sqrt{31}$, not uncertainty from sampling new tasks.

B.2 Matched-margin alternatives

We construct 64 alternative assignments in each of nine analysis cells by permuting reference labels within strata defined by task length, primary prediction and criterion category or scoring mode. This preserves every stratum's prediction/reference confusion counts, the task slots and lengths, and all primary and auxiliary predictions. Root seed 20261007 and a dataset-specific hash define the random streams. Both analytical policies are evaluated at shares $0.05, 0.20, 0.50, 0.84, 0.95$, including originally observed focal and peak regions. An alternative assignment with a different update effect demonstrates that the retained summaries do not determine that effect. These are retrospective counterexamples, not a null test for random reference noise. The permutations can also change reference task pass counts, initial task errors and their joint configuration, so they do not isolate a causal effect of error correlation.

C Certification and repair under different allocations

Let $E_0 = \sum_i \mathbf{1}\{\widehat{Z}_i^0 \neq Z_i\}$. For a query set Q, write $C_W(Q)$ and $C_R(Q)$ for the numbers of certified units that were initially wrong and initially right. Then

$$C(Q) = C_W(Q) + C_R(Q), \tag{19}$$
$$E_G(Q) = E_0 - C_W(Q), \tag{20}$$
$$E_E(Q) = E_0 - C_W(Q) - D(Q) + A(Q). \tag{21}$$

Every certified initially wrong unit is repaired under either rule. The only further eager repairs are the uncertified false-FAIL repairs counted by D, and its only new errors are those counted by A. This proves the last two identities directly from the initial classes.

For two allocations applied to the same frame, define every signed difference as mixed allocation minus uniform criterion sampling. Pairing within a query-order replicate yields

$$\Delta C = \Delta C_W + \Delta C_R, \qquad \Delta E_E = -\Delta C_W - \Delta D + \Delta A, \qquad \Delta E_G = -\Delta C_W. \tag{22}$$

We calculate these differences before averaging, preserving the additive identities in Proposition 3. To describe where additional certificates go, we also report the composition of the *mixed-only* certified set, separately from the uniform-only set. This gross composition remains meaningful when net certification gain is zero or negative. More certificates on initially correct units can provide useful confirmation without repairing an error; it is not by itself evidence that an allocation is harmful.

The fixed mixed design spends up to $\lfloor m/2 \rfloor$ labels on judge-first serial short circuit, then draws a fresh uniform sample without replacement from the remaining criterion slots. If short circuit completes before its phase cap, the unused budget transfers to the random phase. All revealed labels count toward the common actual total m. This contrasts acquisition while the matched-query comparison in Appendix A.3 contrasts updating. Neither design uses hidden initial-error classes to choose queries.

D Finite-population estimation and intervals

D.1 An adaptive first phase followed by fresh sampling

Set $a_u = \mathbf{1}\{P_u = Y_u\}$. Let the first-phase transcript reveal a set C of c criterion labels with known correct count $t_C = \sum_{u \in C} a_u$. Condition on that complete transcript. The remaining $R = M - c$ slots form a fixed population. Draw an independent simple random sample S of size n without replacement from it, and observe $x = \sum_{u \in S} a_u$. For $n > 0$,

$$\widehat{a} = \frac{t_C + Rx/n}{M}, \qquad \mathbb{E}[\widehat{a} \mid C, \text{first-phase observations}] = a. \tag{23}$$

The expectation follows from inclusion probability n/R for every remaining slot. Conditioning on the observed history accommodates the adaptive first phase: the second-phase estimate expands only the unobserved remainder. When $R = 0$, the reference agreement is known exactly. When $R > 0$ but $n = 0$, we report logical bounds $[t_C/M, (t_C + R)/M]$ and no design-unbiased point estimate from this formula.

Let S_R^2 be the finite-population variance of the remainder's agreement bits, with denominator $R - 1$. For $R > 1$ and $n > 0$,

$$\mathrm{Var}(\widehat{a} \mid \text{first phase}) = \frac{R^2}{M^2}\left(1 - \frac{n}{R}\right)\frac{S_R^2}{n}. \tag{24}$$

The factor $1 - n/R$ is the finite-population correction. Census remainders have zero variance. Where a normal approximation is reported, $n \geq 2$ permits the sample variance $s^2 = x(n-x)/[n(n-1)]$ to replace S_R^2. That approximation is evaluated separately from the guaranteed intervals below.

D.2 Inclusive hypergeometric inversion

For unknown remainder success count K, the sample count has probability

$$\Pr_K(X = j) = \frac{\binom{K}{j}\binom{R-K}{n-j}}{\binom{R}{n}}. \tag{25}$$

Retain integer counts consistent with the sample and both inclusive tails:

$$\mathcal{K}_\alpha(x) = \left\{K \in \{x, \dots, R-n+x\} : \Pr_K(X \geq x) \geq \frac{\alpha}{2}, \quad \Pr_K(X \leq x) \geq \frac{\alpha}{2}\right\}. \tag{26}$$

Let K_L and K_U be its minimum and maximum. The microagreement interval is $[(t_C + K_L)/M, (t_C + K_U)/M]$. For every fixed true K, rejection by either tail occurs with probability at most $\alpha/2$; a union bound gives coverage at least $1-\alpha$. Discreteness can make it conservative. This is a conditional, fixed-budget randomization guarantee with the reference held fixed, not simultaneous coverage of every budget or validity under optional stopping. At $n = 0$ the count range is $[0, R]$, and at $n = R$ it is $[x, x]$.

The implementation compares ordinary tails numerically, but recomputes near-threshold decisions using integer binomial coefficients and rational $\alpha/2$. A tolerance triggers exact calculation rather than accepting an endpoint. For example, with $R = 16, n = 2, x = 0, \alpha = 0.05$, $K = 13$ satisfies $\Pr(X = 0) = \binom{3}{2}/\binom{16}{2} = 0.025$ exactly and must remain in the interval $[0, 13]$. This resolves the floating-point boundary case without altering the mathematical acceptance rule.

D.3 Prediction strata and a charged pilot

Prediction strata are the nonempty sets with $P_u = 0$ or $P_u = 1$, known before reference acquisition. Proportional allocation reserves one draw per nonempty stratum when affordable, distributes remaining draws proportional to stratum size, respects capacities, and resolves integer remainders by descending fractional part with prediction-zero first on ties. Otherwise it falls back to uniform sampling.

For the charged-pilot design, the initial allocation is $\min(2, N_h)$ labels in stratum h. The design reserves at least one later draw in each stratum not exhausted by that initial allocation. If the cap cannot fund both stages, it falls back to uniform sampling. Otherwise the pilot target is the larger of this minimum and $\lfloor 0.2m \rfloor$, capped to preserve the later draws. Additional pilot slots are allocated by stratum size, leaving an unobserved slot in each initially noncensused stratum. After pilot counts p_h and observed agreement totals t_h, define

$$\widetilde{a}_h = \frac{t_h + 1/2}{p_h + 1}, \qquad R_h = N_h - p_h, \qquad w_h = R_h\sqrt{\widetilde{a}_h(1 - \widetilde{a}_h)}.$$

The remaining budget is allocated by weights w_h, with a minimum of one draw from each nonempty remainder and its capacity as an upper bound. The same capacity-constrained proportional allocation and largest-remainder rule determine integer counts. Pilot and remainder samples use separate random-priority streams. All pilot labels are charged to the budget.

Conditional on the pilot, let the remainder sample in stratum h have size n_h and agreement total x_h. Then

$$\widehat{a}_{\text{strat}} = \frac{\sum_h t_h + \sum_{h:R_h>0} R_h x_h/n_h}{M} \tag{27}$$

is conditionally unbiased. If H strata are not censused, we obtain a count interval in each at error level α/H, sum their lower and upper bounds and add the known pilot total. The union bound proves conditional coverage at least $1 - \alpha$ for the overall mean; independence between stratum interval events is not required. Censused strata contribute their exact counts. The accompanying normal approximation sums the variances in (24), stratum by stratum, and clips endpoints to $[0, 1]$. It is unavailable when a noncensused stratum has fewer than two sampled slots. These design-based controls are not an implementation of StratPPI.

At each of six shares $0.006, 0.05, 0.20, 0.50, 0.60, 1$, we report bias, empirical RMSE, coverage and interval width over 31 query-order replicates in each of nine analysis cells. RMSE is $[31^{-1}\sum_{s=1}^{31}(\widehat{a}_s - a)^2]^{1/2}$. The replications describe the finite frames; coverage guarantees derive from the sampling design and inversion, not the number of observed covered replicates. Conservative simultaneous stratum bounds can be wider despite a smaller empirical RMSE.

D.4 Whole-task sampling

For estimating θ, the separate whole-task panel fixes the number of sampled evaluation units in advance and draws them uniformly without replacement. Fully revealing an included unit and short-circuiting after its first reference failure yield the same certified Z_i. Their sample means and finite-population task-count intervals are consequently identical, although their criterion-query costs differ. Here sample size is fixed and label cost is random. This experiment is distinct from stopping task sampling at a fixed label cap, which would generally alter inclusion probabilities and require its own inference.

E Alternative references and task composition

E.1 Matched reference panels

The matched frame contains 206 outputs with at least two distinct raters. The corresponding 431 annotations are distinguished from the original 1,539 annotation units. Outputs are sorted by base-task and output identity. For each output, distinct-rater annotations are sorted by the SHA256 digest of the literal string `reference-panel-v1|output|rater|annotation`; the first two complete rubric vectors form panels A and B. A separate ordering with salt `judge-anchor-v1` chooses the annotation supplying the saved primary and auxiliary predictions. The selection rule does not inspect label values or agreement. Rubric-slot identities are checked before alignment, and the same selections are used for both targets and both judges. A and B thus refer to symmetric selections across outputs, not globally ranked raters or a chronological improvement in reference quality.

Crossed comparisons distinguish the reference used for acquisition from the reference used for reading and evaluating the resulting query set. For acquisition panel $a \in \{A, B\}$, an adaptive policy obtains $Q^{(a)}$ using that panel's answers. For evaluation panel $e \in \{A, B\}$, queried slots are re-read using $Y^{(e)}$, both updates are computed from those answers, and residual disagreement is measured against the complete $Y^{(e)}$. Off-diagonal comparisons therefore describe counterfactual re-reading of fixed queried items. They do not retain panel A answers while silently treating panel B as truth. Nonadaptive policies have identical query sets for either acquisition panel. Adaptive short circuit can change its stopping decisions and actual query count.

The update grid includes uniform criterion, disagreement-then-random and judge-first short circuit at all 101 budget shares and seeds 1–31. The allocation grid compares the mixed and uniform designs at the six shares listed above, with equal actual query counts within each contrast. The four crossed means share outputs and references and are not four independent replications. Moving from the original frame to the matched frame also changes composition, weighting and judge anchoring; only within-frame, fixed-prediction contrasts isolate the reference-panel change.

E.2 Base-task composition replay

For the two JudgmentBench GPT-5.4 targets and DR Gemini, we generate 399 bootstrap compositions with root seed 20261008. Each replicate draws the original number of base tasks with replacement and retains every output, rater and criterion belonging to a selected task. The two JudgmentBench targets share base-task multiplicities; DR has a separate random stream. Original release-relative unit order is preserved by visiting original units and cloning each according to its base-task multiplicity. Shared base-task/copy identifiers retain dependence within a replicated task. Each composition gets its own N, M and recomputed rounded budget caps.

For updating, every composition uses the exact order expectations in Appendix B for both analytical policies over the full 101-point grid. For allocation, each composition reruns the 20% mixed versus uniform contrast using the same fixed 31-seed panel, taking paired differences before their mean. Thus the latter ranges include sensitivity of a fixed finite Monte Carlo panel, rather than separately estimating its Monte Carlo error. Quantiles use linear interpolation. We show pointwise 2.5th and 97.5th percentiles in counts and per 100 evaluation units, and retain each replicate's recomputed peak and its location. The archived approximate simultaneous construction uses the 95th percentile of maximum absolute grid deviations; it is omitted from the main figure because its constant width obscures the known zero effects at the two endpoints.

This replay evaluates sensitivity to empirical task composition. In JudgmentBench, 30 curated base tasks do not establish an exchangeable sample from a new task population. Raters remain fixed, and reference-panel and composition experiments are conducted separately. Their ranges therefore do not estimate joint uncertainty over tasks and raters or correct selection bias. A pointwise range at the original curve's selected peak is also not a simultaneous guarantee for that curve.

F Information disclosure and sharp logical endpoints

F.1 Query-only identification

With only queried labels and no restrictions linking unobserved values, let $n_P = \sum_i C_i^+$ and $n_F = \sum_i C_i^-$. Every uncertified unit has at least one unobserved slot and no observed failure. Setting all its unobserved slots to one makes it PASS; setting any one of them to zero makes it FAIL. Because the unrestricted completions can be chosen independently by unit, the sharp endpoint range is

$$\theta \in \left[\frac{n_P}{N}, 1 - \frac{n_F}{N}\right], \qquad \text{width} = 1 - \frac{n_P + n_F}{N}. \tag{28}$$

Attainable finite-population values have increments $1/N$; the displayed interval is their endpoint hull. Its width is a logical consequence of missing information, not a sampling confidence interval. A random sample can estimate the mean precisely without certifying every population unit.

F.2 Fixed references with withheld associations

The disclosure experiment keeps the complete observed labels fixed when constructing a release and changes what the reconstruction procedure may see. Visible information includes task slots and lengths, criterion categories, and the primary predictions. D0 supplies the exact number of reference PASS bits within each task-length-by-prediction group. D1 refines those groups by criterion category. D2 additionally restores the actual reference values attached to all slots of selected whole tasks. These are reference-containing count summaries and restored links, not additional label acquisition.

Let $\mathcal{B}_g$ partition the slots into disclosed groups, with reference success count t_g. Unknown binary assignments x_u and task conjunctions z_i must satisfy

$$x_u, z_i \in \{0, 1\}, \tag{29}$$

$$z_i \le x_u \qquad (u \in \mathcal{J}_i), \tag{30}$$

$$\sum_{u \in \mathcal{J}_i} x_u - z_i \le k_i - 1 \qquad (i = 1, \ldots, N), \tag{31}$$

$$\sum_{u \in \mathcal{B}_g} x_u = t_g \qquad \text{for every disclosed group}, \tag{32}$$

$$x_u = Y_u \qquad \text{for restored slots}. \tag{33}$$

The two conjunction inequalities enforce $z_i = \prod_{u \in \mathcal{J}_i} x_u$: any zero slot forces $z_i = 0$, and all one slots force $z_i = 1$. Minimizing and maximizing $N^{-1} \sum_i z_i$ gives the sharp finite feasible endpoints. This is a binary mixed-integer program, not a continuous linear-program relaxation. Every feasible binary assignment is an allowed completion and vice versa, proving sharpness of the optima in this information model. Extra disclosures can only remove completions, and restoring all labels recovers the observed score.

The program separates exactly by task length because every disclosed group lies within one length stratum. Each endpoint has an attained integer witness, independently checked equality and conjunction constraints, and a closed solver objective bound. The solver receives only the constructed release; hidden labels are used by the publisher to form summaries and later to check containment and full recovery. The experiment restores nested whole-task prefixes at fractions $0, 0.1, 0.25, 0.5, 0.75, 1$, with $\lfloor fN \rfloor$ tasks, under ten saved random orders per population and root seed 20260907. The four RuVerBench runs are analyzed separately. Disclosure-order variation is not sampling uncertainty.

D0 already determines the task-uniform mean-criteria reference score:

$$\mu = \frac{1}{N} \sum_{i} \frac{1}{k_i} \sum_{u \in \mathcal{J}_i} Y_u = \frac{1}{N} \sum_{g} \frac{t_g}{k_g}, \tag{34}$$

where k_g is the common task length in group g. It need not determine the nonlinear conjunction. Because the predicted task score is visible, subtracting it from each reference endpoint gives the feasible correction range. Its sign concerns correction of one judge's score on fixed outputs, not ranking two solver models. A disclosed reference task-pass total would also identify the aggregate mean without revealing every criterion link; the studied disclosure ladder is one sufficient-information comparison, not a universally minimal release scheme.

F.3 A complete two-task example

Take two tasks with two binary slots each. With no queries or extra information, their conjunction mean lies between zero and one. A trusted total of exactly two PASS bits reduces the feasible means to zero or $1/2$: assignments $(1,0),(1,0)$ and $(1,1),(0,0)$ attain the two endpoints. If both first-task slots are then observed as PASS, the trusted total forces both second-task slots to FAIL, identifying the mean as $1/2$. Without that total, the same two observations leave a range $[1/2, 1]$. Thus a task can be unqueried yet logically determined by additional information. Conversely, knowing only that exactly one of the two tasks passes identifies the mean as $1/2$ without identifying which task passes. These examples distinguish label acquisition, individual identification and aggregate identification. The real RuVerBench release contains the task links; the experiment investigates what its coarsened summaries support, rather than alleging a missing-link defect in that release.

G Additional displays and reproduction

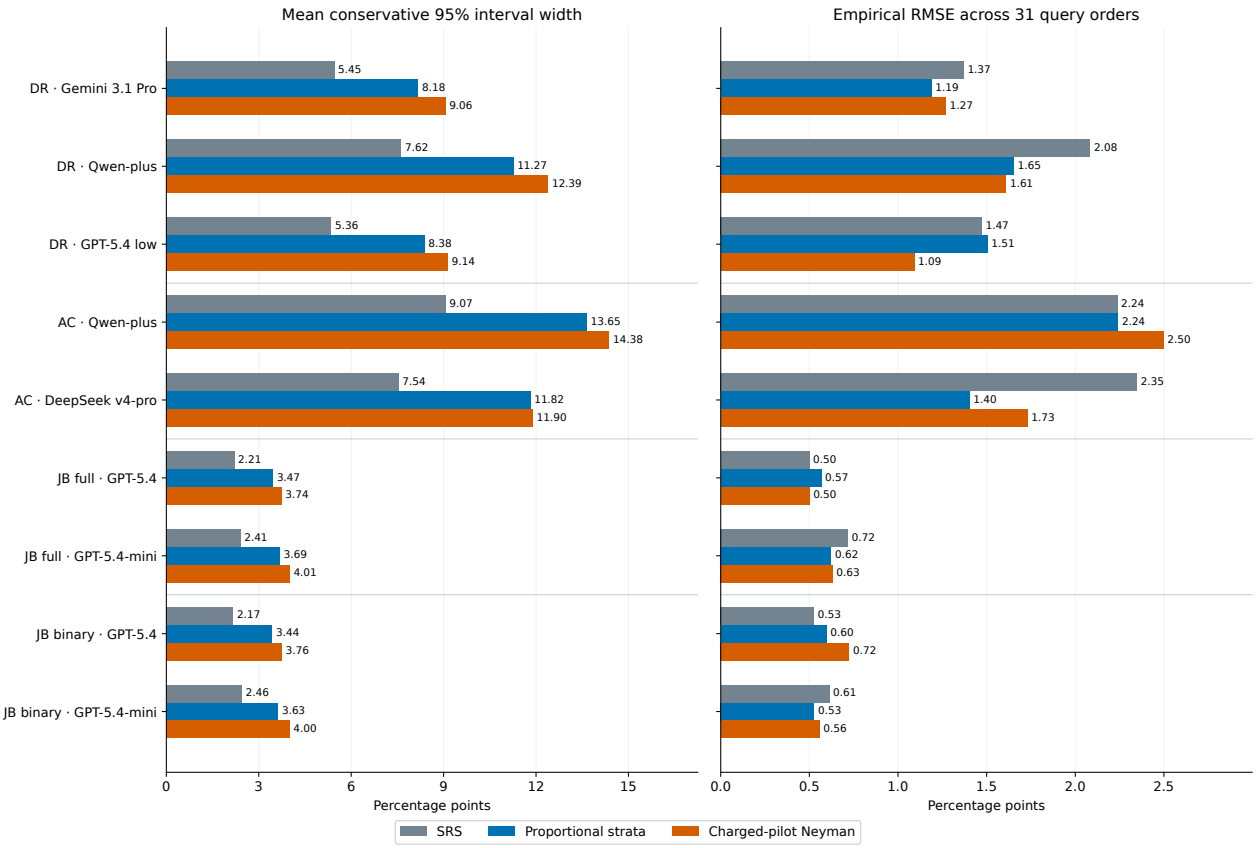


Figure 3: Estimation controls at a 20% criterion budget across all nine cells. Guaranteed interval widths and empirical RMSE are shown separately. The stratified intervals combine finite-population count bounds conservatively; their width alone does not rank point-estimation performance. Pilot labels count toward the same budget. Each empirical quantity summarizes 31 saved orders, while the coverage guarantee follows from the design.

The released package separates data integrity, scientific replay, and manuscript validation. Inputs are compact numerical projections with task, output, rater, criterion and scoring metadata. Original input provenance and content hashes are in `data/PROVENANCE.json`; result provenance is stored with the archived numerical outputs. The following commands run from the repository root after installing the pinned Python dependencies:

```
python -m src.check_package
python -m pytest -q
python -m src.reproduce --mode full --output artifacts/reproduced
python -m src.verify_release --results artifacts/reproduced
python -m src.reproduce_extensions --output artifacts/reproduced-extensions
python -m src.reproduce_extensions --output artifacts/reproduced-extensions --verify-only
python -m src.verify_manuscript_claims --output artifacts/paper-checks/claims.json
python -m src.build_paper
```

The final command additionally requires Tectonic (tested with version 0.17.0) and writes a rebuilt PDF separately from the published copy. Initial dependency installation can require network access; the scientific replay does not call judge APIs. Rebuilding the PDF can change PDF metadata without changing the scientific results. The published manuscript includes a claim audit and a citation audit; the broader technical reading library remains separate from the selected manuscript bibliography.

Table 4 maps the paper's numerical claims to the result groups. For compactness, paths below are relative to `artifacts/expected/`. The executable claim audit records exact row keys, units, recomputed values and source hashes. It checks selected paper anchors and accounting identities; the complete scientific reproduction commands compare the larger archived grids.

Table 4: Numerical source routing for the paper.

Paper component	Archived source group
Frames and initial errors	`conjunction_policy_study/study.json` and input tables
Query-order curves	`conjunction_policy_study/budget_curves.csv.gz`
Exact expectations and permutations	`conjunction_mechanism/`
Paired allocation decomposition	`conjunction_mechanism/`
SRS/mix and whole-task estimates	`conjunction_label_value/`
Reference and composition sensitivity	`conjunction_robustness/`
Stratified estimation controls	`conjunction_robustness/`
Disclosure endpoints and witnesses	`information_disclosure_replay.json`